# SkyPredict AI

Technical Questions & Interview Preparation

Prepared for mentor presentation and technical interviews. The questions are grounded in the current project structure and documented workflows.

## 1. Project Architecture

**Q. What is SkyPredict AI?**

Answer: An end-to-end Streamlit flight analytics application covering flight-price prediction, passenger-satisfaction prediction, route analytics, dashboards and model inference.

**Q. What technologies are used?**

Answer: Python, Pandas, NumPy, Streamlit, Scikit-Learn, Plotly and Joblib.

**Q. What is the application architecture?**

Answer: Datasets and serialized models are loaded first; utility and feature-building functions prepare inputs; Streamlit renders the UI; the selected page performs analysis or model inference.

**Q. Why Streamlit?**

Answer: It allows an ML project to become an interactive Python web application without a separate frontend framework.

## 2. Flight Price Prediction

**Q. Is flight-price prediction regression or classification?**

Answer: Regression, because the output is a continuous ticket price.

**Q. What happens from search to prediction?**

Answer: The user selects Source, Destination, Journey Date and Class; historical route records are displayed; a selected flight is stored in session state; the Prediction page builds model-compatible input and calls flight_model.predict().

**Q. Why is Journey Date not used to filter the historical CSV?**

Answer: The CSV is treated as historical training data. A future customer date may not exist there, so the date is used as a model input rather than as a historical availability filter.

**Q. What is feature engineering in this project?**

Answer: Departure time is split into hour/minute, duration is converted to minutes, stops are converted to a number, and journey day/month are derived from the selected date.

**Q. How is the predicted price interpreted?**

Answer: The UI categorizes the predicted amount into Budget, Average or Premium fare bands. These thresholds are application-level interpretation, not another ML model.

## 3. ColumnTransformer / ML Error Debugging

**Q. Why did the TypeError 'NoneType' object is not iterable occur?**

Answer: The error occurs inside Scikit-Learn ColumnTransformer.transform() when the fitted transformer compares its expected training columns with the columns supplied at prediction time. It indicates an input-schema mismatch or invalid fitted transformer metadata.

**Q. How would you debug it?**

Answer: Inspect the model type, named_steps, preprocessing step, feature_names_in_, transformer column definitions and the final DataFrame columns. Compare expected versus actual schema before calling predict().

**Q. Why should you not blindly add columns?**

Answer: The pipeline may already perform encoding or transformation. The fitted pipeline is the source of truth for what raw input it expects.

**Q. What is the safest prediction sequence?**

Answer: Validate inputs → inspect fitted schema → construct one-row DataFrame → match expected columns → call predict() inside try/except → render the result only if prediction succeeds.

## 4. Customer Satisfaction

**Q. Is customer satisfaction classification?**

Answer: Yes. It predicts a satisfaction class such as satisfied/not satisfied.

**Q. Should the customer enter Flight Distance?**

Answer: No. The application is designed to determine route distance automatically where possible.

**Q. How is distance calculated?**

Answer: The application can derive it from route/dataset information and, when coordinates are available, calculate approximate geographic distance using the Haversine formula.

**Q. Why is automatic distance better?**

Answer: It removes an unnecessary technical input from the customer and reduces manual entry errors.

**Q. How is confidence calculated?**

Answer: If the classifier supports predict_proba(), the maximum class probability is displayed as a percentage.

## 5. Pandas & Data Processing

**Q. Why Pandas?**

Answer: For CSV loading, filtering, grouping, aggregation, feature preparation and DataFrame construction.

**Q. How are routes counted?**

Answer: Source and Destination pairs are deduplicated and counted.

**Q. How is average fare calculated?**

Answer: Price is converted to numeric values and averaged while handling invalid values.

**Q. Why convert duration to minutes?**

Answer: It creates a numeric representation suitable for analysis and potential model features.

## 6. Dashboard & Plotly

**Q. What does the dashboard show?**

Answer: Flight count, airline count, route count, average fare, airline prices, route prices, stop distribution, duration-price relationships, monthly trends and airline comparison.

**Q. Why Plotly?**

Answer: It provides interactive browser-based charts.

**Q. What does groupby() do?**

Answer: It groups records by fields such as Airline or Route_Name so statistics such as mean, minimum and maximum can be calculated.

**Q. Does Duration vs Price prove causation?**

Answer: No. It only shows an observed relationship in the dataset; correlation does not establish causation.

## 7. Streamlit Engineering

**Q. Why session_state?**

Answer: Streamlit reruns the script after interactions. session_state preserves the selected flight and navigation state across those reruns.

**Q. Why URL query parameters?**

Answer: They provide an additional way to preserve and resolve the current page.

**Q. Why caching?**

Answer: Caching avoids repeatedly loading the same CSVs and model resources during reruns.

### Q. How does navigation work?

Answer: Sidebar buttons update session state and the page query parameter, then rerun the application.

## 8. Error Handling

### Q. How are missing datasets handled?

Answer: The loader returns an empty DataFrame and the relevant page displays an explanatory error.

### Q. How are missing models handled?

Answer: The page checks for None before prediction.

### Q. Why did NameError: level occur?

Answer: If model prediction failed before level was assigned, later UI code referenced an undefined variable. Prediction-dependent variables should be initialized before inference and rendered only after successful prediction.

### Q. What is robust error handling for prediction?

Answer: Validate schema, run prediction inside try/except, maintain an explicit success flag, and avoid using prediction outputs when inference failed.

## 9. ML Concepts to Know

### Q. What is a Pipeline?

Answer: A Scikit-Learn object that bundles preprocessing and an estimator so inference uses the same transformation sequence.

### Q. What is ColumnTransformer?

Answer: It applies different transformations to selected input columns.

### Q. What is feature engineering?

Answer: Transforming raw application data into variables appropriate for model inference.

### Q. What is one-hot encoding?

Answer: Representing categorical values with binary indicator features.

### Q. What is predict_proba()?

Answer: A classifier method that returns class probabilities when supported.

### Q. What is Haversine distance?

Answer: An approximation of great-circle distance between two latitude/longitude points.

## 10. Difficult Interview Questions

### Q. Why Random Forest?

Answer: If the saved model is confirmed to be Random Forest, explain that it can model nonlinear relationships and feature interactions and is relatively robust. Verify the actual estimator inside the .pkl before claiming this.

### Q. What metrics should you know?

Answer: For regression, know MAE, RMSE and $R^2$. For classification, know accuracy, precision, recall, F1 and ROC-AUC. Use the actual training metrics from your model-training work rather than inventing them.

### Q. How did you prevent data leakage?

Answer: Only claim the exact leakage-prevention process used during training. The Streamlit app itself is mainly an inference layer.

### Q. How would you improve the project?

Answer: Separate UI, data access, feature engineering and inference into modules; add tests; validate schemas at startup; version models; improve route-distance coverage; add model-performance reporting; and hide developer debug output in production.

### Q. How would you deploy it?

Answer: Package the Streamlit application with dependencies, datasets and model artifacts, then deploy to a suitable hosting environment. Production deployment should add logging, model versioning, validation and monitoring.

## 11. 60-Second Presentation

“SkyPredict AI is an end-to-end flight analytics application built with Python and Streamlit. It has two ML components: flight-price regression and passenger-satisfaction classification. Pandas handles data processing, Joblib loads trained model artifacts, Scikit-Learn performs inference and Plotly provides interactive analytics. For flight prediction, the application takes route, date and flight information, engineers the required features and prepares a DataFrame matching the trained model schema before prediction. For satisfaction prediction, the application collects passenger and service information and determines flight distance automatically where possible. A major engineering focus was handling Streamlit reruns and ML pipeline schema compatibility.”

## 12. Mentor Questions You Should Be Ready to Answer

• Why did you choose this problem?

• What was the original dataset and target variable?

• What preprocessing was performed during training?

• Which exact features were used by each model?

• Which algorithm and hyperparameters were finally selected?

• What validation strategy was used?

• What were the final RMSE/$R^2$ and classification metrics?

• How did you handle missing values and categorical variables?

• How did you ensure inference used the same preprocessing as training?

• What caused the ColumnTransformer error and how did you debug it?

• How is flight distance obtained without asking the customer?

• What would you change before deploying this to real customers?

## 13. Final Interview Checklist

■ Know the business problem in one sentence.

■ Know why price is regression and satisfaction is classification.

■ Know the exact target column for each model.

■ Know the exact training features and preprocessing.

■ Know Pipeline and ColumnTransformer.

■ Know the cause of the earlier schema-mismatch error.

■ Know the Home → session_state → Prediction flow.

■ Know how automatic route distance works.

■ Verify the actual estimator inside each .pkl before saying Random Forest.

■ Know the real training metrics from the model-training work.

■ Be able to explain one limitation and three future improvements.